

Notes on The Implementation and Backtesting of Diffusion Model for Prediction Market Probabilities Utilizing Polymarket API Pricing Data

Dylan Gabriel

1 Introduction

Prediction market contracts are often quoted as prices between 0 and 1, where the price can be interpreted as a market-implied probability of an event occurring. Modeling randomness with a standard arithmetic Brownian motion introduces bounding issues. My goal within this research was to explore a simple stochastic model bounded between 0 and 1 for prediction market probabilities. I then calibrated the model using historical market data and compared the simulated probabilities/prices to the actual market probabilities/prices.

Let $P_t \in [0, 1]$ denote the prediction market probability at time t . I model P_t using the Stratonovich stochastic differential equation

$$dP_t = \alpha \sqrt{P_t(1 - P_t)} \circ dW_t,$$

where $\alpha > 0$ controls the volatility of belief changes in the market and W_t is a standard Brownian motion. The factor

$$\sqrt{P_t(1 - P_t)}$$

is maximized at $P_t = 1/2$ and shrinks near 0 and 1. This makes sense since volatility should increase during periods of uncertainty.

2 Transformation

Under the Stratonovich interpretation, the usual chain rule applies. Separating variables gives

$$\frac{dP_t}{\sqrt{P_t(1 - P_t)}} = \alpha \circ dW_t.$$

Since

$$\int \frac{dP}{\sqrt{P(1 - P)}} = 2 \arcsin(\sqrt{P}),$$

we obtain

$$2 \arcsin(\sqrt{P_t}) = 2 \arcsin(\sqrt{P_0}) + \alpha W_t.$$

Absorbing the factor of 1/2 into the volatility parameter, the solution can be written as

$$\boxed{P_t = \sin^2 \left(\arcsin(\sqrt{P_0}) + \alpha W_t \right)}.$$

This gives a closed-form bounded model for prediction market probabilities.

3 Calibration

Given observed prices $P_{t_0}, P_{t_1}, \dots, P_{t_n}$, I use the transformation

$$Y_{t_i} = \arcsin(\sqrt{P_{t_i}}).$$

Then the increments are

$$\Delta Y_i = Y_{t_i} - Y_{t_{i-1}}.$$

For the model

$$dY_t = \alpha dW_t,$$

we have approximately

$$\Delta Y_i \sim N(0, \alpha^2 \Delta t_i).$$

Therefore, a natural estimator is

$$\hat{\alpha} = \sqrt{\frac{1}{n} \sum_{i=1}^n \frac{(\Delta Y_i)^2}{\Delta t_i}}.$$

In this implementation, market price history was pulled from the Polymarket API for the market “Will Bitcoin hit \$1m before GTA VI?” Using pandas dataframes, I implemented the simulation and calibration procedures.

4 Simulation

After estimating α , future probability paths can be simulated directly. Starting from a current probability P_0 , define

$$Y_0 = \arcsin(\sqrt{P_0}).$$

For simulated Brownian paths W_t , the prediction probability is

$$P_t = \sin^2(Y_0 + \hat{\alpha} W_t).$$

This is useful because the simulated paths automatically remain in $[0, 1]$.

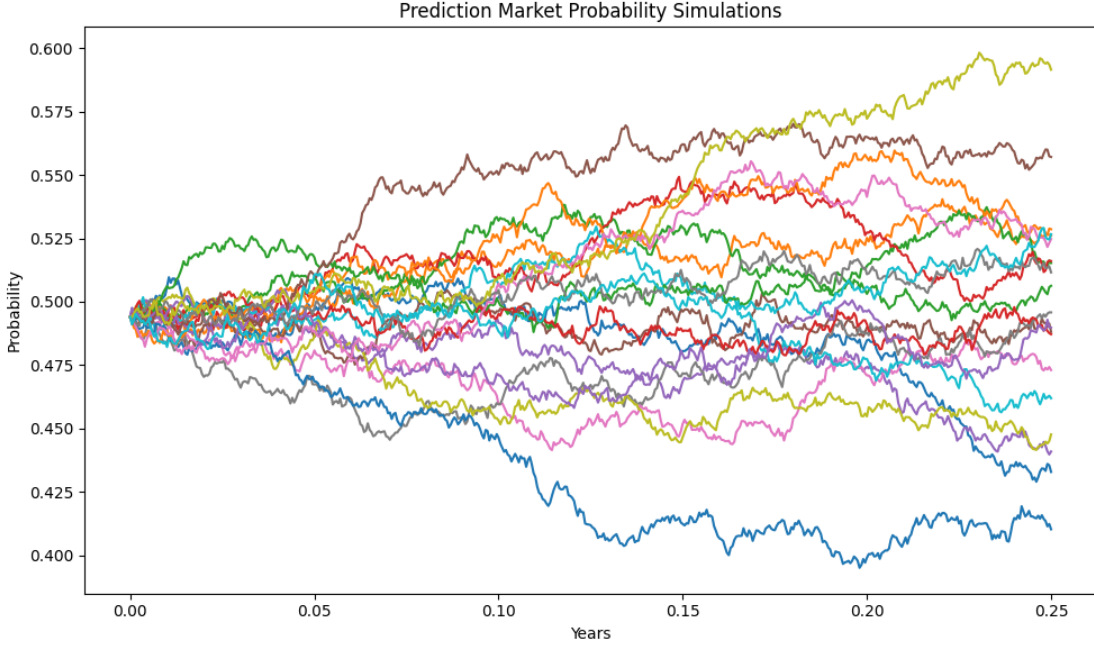


Figure 1: Simulated prediction market probability paths under the angular diffusion model.

5 Backtesting

To test the model, I split the historical data into a training set and a test set. The training set was used to estimate α . Starting from the last observed training price, I simulated many paths over the same time window as the test set. I then compared the realized test path against the simulated confidence bands. This is useful for determining the stability of the model and observing how well the confidence bands represent actual market behavior.

Let P_t^{realized} denote the realized market probability. A simple backtesting procedure can then be used to check whether the realized path stays inside the simulated 90% confidence band.

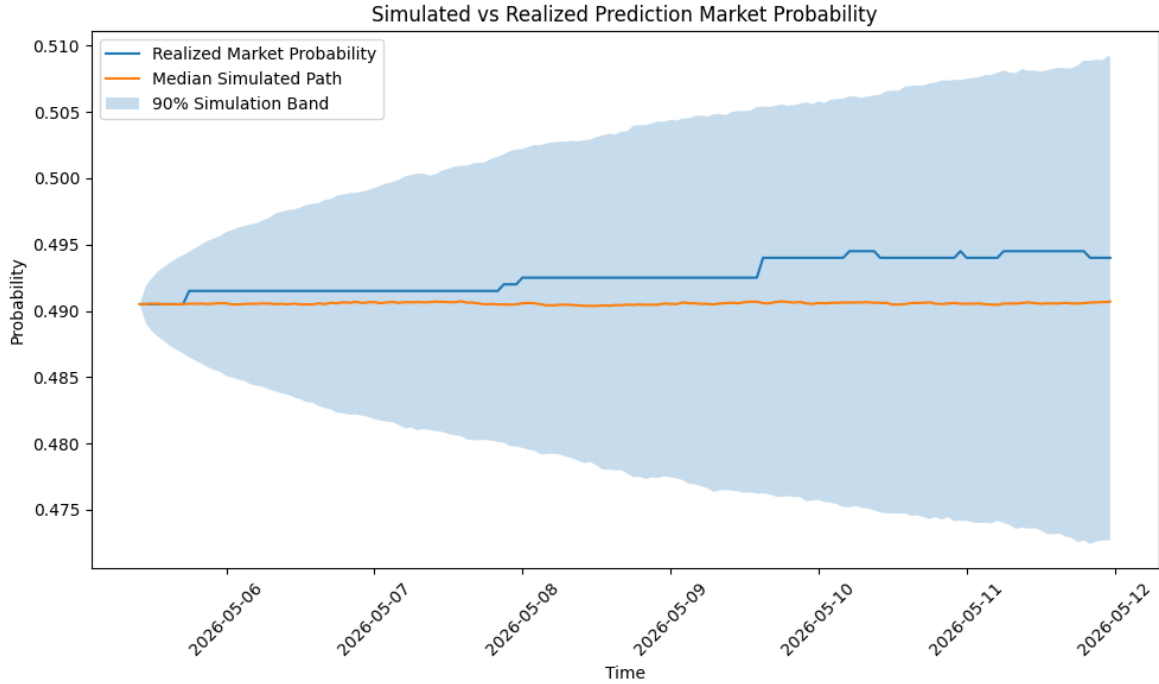


Figure 2: Simulated versus actual prediction market probabilities. The blue line represents the realized market probabilities, the orange line represents the mean simulated path, and the shaded region is the 90% simulation band.

The realized path remaining inside the simulation band suggests that the observed market movement was statistically plausible under the angular diffusion model. The widening of the band over time is also consistent with Brownian scaling, since uncertainty grows proportionally to $\sqrt{t}$.

6 Discussion

The model appears to work reasonably well. It is useful that the model itself remains bounded between 0 and 1. The stochastic differential equation also has a closed-form solution, which makes both simulation and calibration significantly easier.

There are also real limitations. Prediction markets often move discontinuously after news, announcements, or information shocks. A pure diffusion model may not capture these sudden jumps. Adding a jump process to the model may help remedy this and improve the predictive power of the model. Additionally, near resolution, the market typically converges toward 0 or 1 as participants become increasingly certain whether an event will or will not occur. This may require a time-dependent volatility term rather than the constant volatility used in this simpler exercise. Still, as a first model, the diffusion framework provides a useful foundation for further study of prediction markets.

7 Possible Extensions

Several extensions are natural:

$$dY_t = \mu_t dt + \alpha_t dW_t,$$

where μ_t captures directional belief drift and α_t allows volatility to change over time. Another possible extension is a jump-diffusion model,

$$dY_t = \mu dt + \alpha dW_t + J dN_t,$$

where N_t is a Poisson process and J represents a news shock. However, the pure diffusion model is simpler and preserves the clean closed-form structure.

8 Conclusion

These notes developed a bounded stochastic model for prediction market probabilities using the transformation

$$Y_t = \arcsin(\sqrt{P_t}).$$

Under the Stratonovich interpretation, this transformation linearizes the dynamics, giving Brownian motion in angular probability coordinates. The resulting model can be calibrated using real market data, simulated in closed form, and backtested against realized probability paths. This provides a simple but useful framework for modeling prediction market probabilities.